

# 21CSC529T Inferential Statistics

## Unit - 3: Estimation & Hypothesis Testing

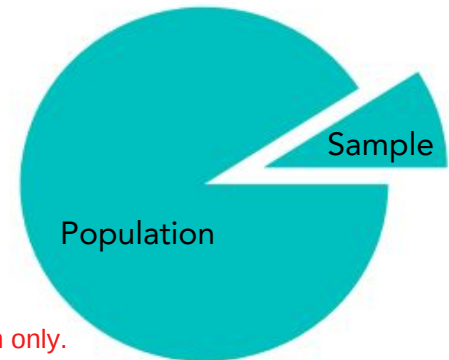
## Unit 3: Estimation & Hypothesis Testing

- Sampling Distribution & Central limit theorem,
- Point estimate
- confidence interval
- Hypothesis Testing Basics
- Null & Alternative Hypothesis formulation - Types of hypotheses
- Hypothesis testing process
- Errors in hypothesis testing - Type I and Type II
- Power of test
- Number of tails
- Choice of test statistic

# Population and Sample

# Population & Sample

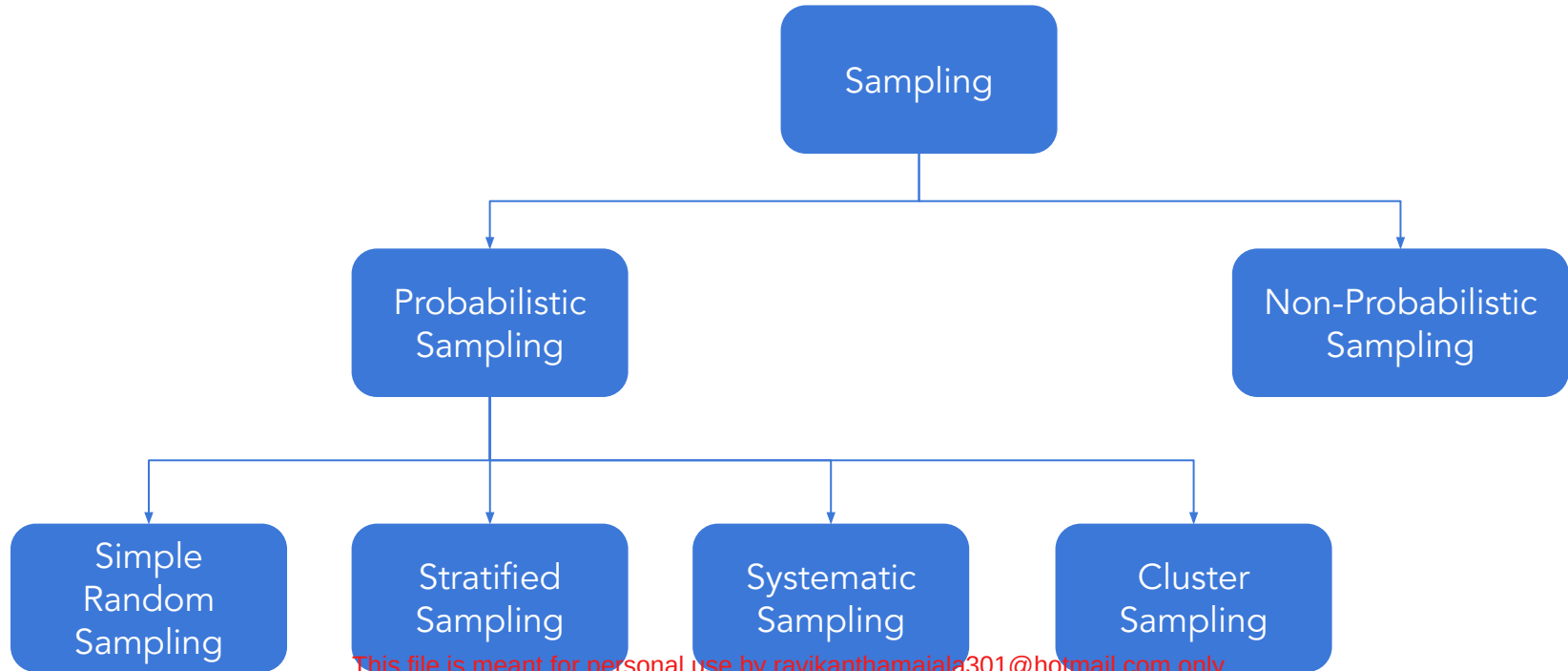
- **Population** is a collection of all the individuals or objects
- **Sample** is a subset of the population which is a representative of the population



# Need to draw a sample

- In some of the real case scenarios we may not be able to gather all the observations in the population
- In such cases we can consider a sample which can be used to draw inferences about the population
- Sampling can reduce the computational time and cost
- In machine learning models, a sample obtained using an appropriate sampling strategy can provide accurate results with the less number of observations

# Sampling techniques



This file is meant for personal use by ravikanthamajala301@hotmail.com only.

Sharing or publishing the contents in part or full is liable for legal action.

# Simple random sampling (SRS)

- Simplest and most common method
- Sample is drawn unit by unit
- Each unit has equal probability of being selected
- Two ways of selecting units:
  - Without replacement: if the selected unit is not returned to the population
  - With replacement: if the selected unit is returned to the population

# SRS without replacement

- At any specific stage in the sampling, the probability of any of the remaining units is  $1/N$ , where  $N$  is the total number of sample units
- From  $N$  sample units, if a sample of size  $n$  is drawn, there are  ${}^N C_n$  possible samples
- The probability of selecting one of them is  $1/{}^N C_n$



# SRS without replacement



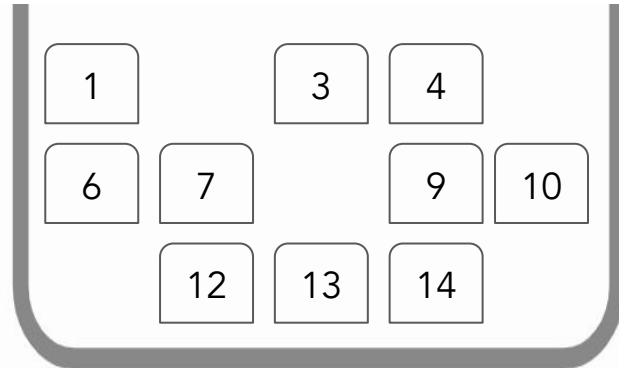
The unit selected to be in the **sample** is not returned to the **population**.



# SRS without replacement - example

Drawing a sample from manufactured articles to know the proportion of defectives.

Manufactured articles



Sample to test for defective articles



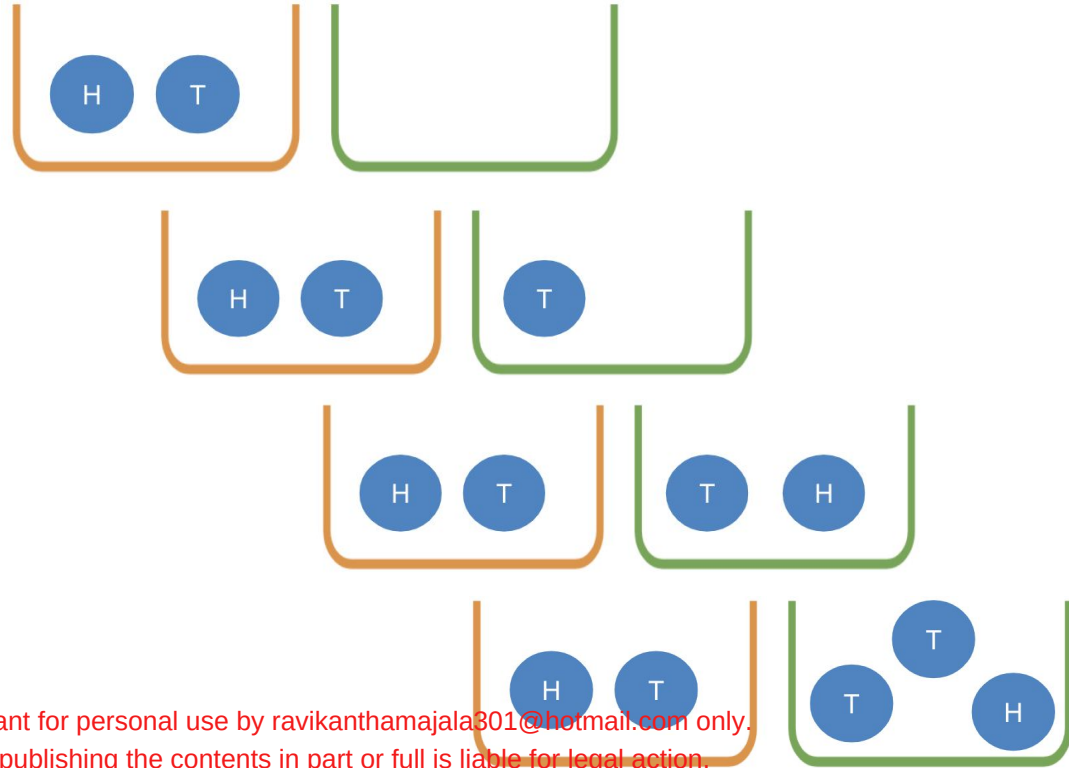
This file is meant for personal use by [ravikanthamajala301@hotmail.com](mailto:ravikanthamajala301@hotmail.com) only.

Sharing or publishing the contents in part or full is liable for legal action.

# SRS with replacement

- The probability of selecting a specific unit at any given draw is always  $1/N$ , where  $N$  is total number of sample units
- From  $N$  sample units, if sample of size  $n$  is drawn, there are  $N^n$  possible samples
- The probability of selecting one of them is  $1/N^n$
- Example: You have 5 fruits and want to take a sample of 2. You can choose a fruit more than once.

# SRS with replacement



The unit selected to be in the **sample** is returned to the **population**.

# SRS with replacement - example

- Generally used in simulation studies
- A car insurance company wants to set the premium amount. The probabilities for major and minor accident are known to be 0.02% and 2% respectively  
(assume that an insuree can meet with an accident only once)
- A SRS with replacement is conducted to simulate 1 million such scenarios

# Simple random sample - python function

| Python function               | Description                     |
|-------------------------------|---------------------------------|
| <code>random.choices()</code> | draw sample with replacement    |
| <code>random.sample()</code>  | draw sample without replacement |

# Summary

| Sampling Method | Summary                                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Simple Random   | Mostly used method. The sample can be selected with or without replacement.                                                               |
| Stratified      | Sample point is selected from each subgroup in the population.                                                                            |
| Systematic      | Easy to implement for large population. The first sample point is selected randomly and remaining points are selected at fixed intervals. |
| Cluster         | An entire subgroup in the population is selected as a sample point.                                                                       |

# Summary

- Sample is the subset of the population
- Different sampling methods: SRS with replacement (SRSWR), SRS without replacement (SRSWOR), Stratified, Systematic, Cluster sampling
- SRSWR method can be used if one wish to select the sample observation more than once
- Stratified sampling can be used when data is heterogeneous
- Cluster sampling can be used when the data consists of several groups



# Resampling

# Resampling

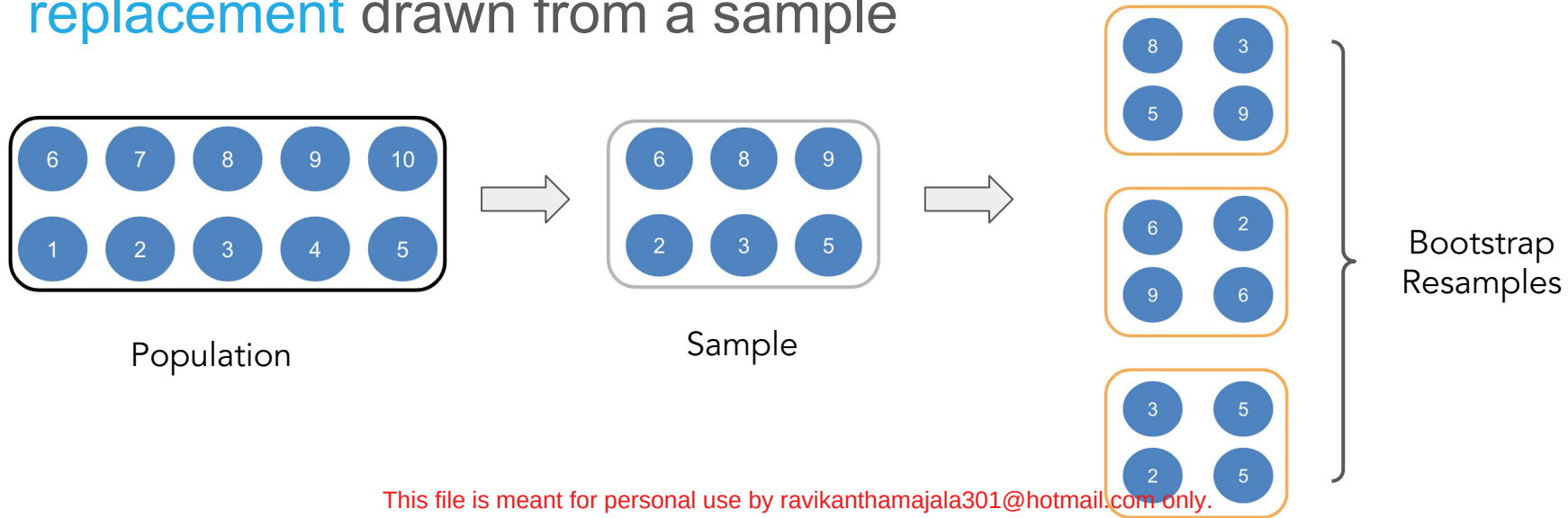
- Resampling is a technique in which we draw a sample from the available sample
- Resampling often involves performing similar testing methods with sample sizes within that group.
- This can mean testing the same sample, or reselecting samples that can provide more information about a population.
- The resampling techniques are:
  - Bootstrap resampling
  - Jackknife resampling

This file is meant for personal use by ravikanthamajala301@hotmail.com only.

Sharing or publishing the contents in part or full is liable for legal action.

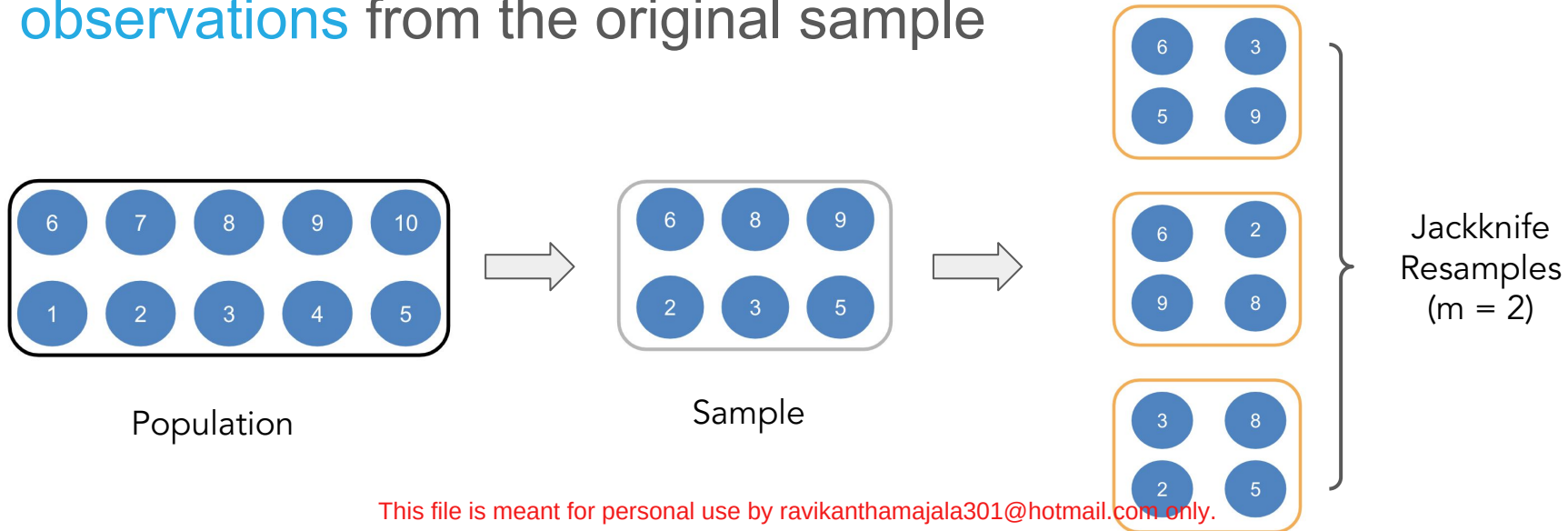
# Bootstrap resample

Bootstrap resample is a **simple random sample with replacement** drawn from a sample



# Jackknife resample

Jackknife resample is a method in which one **deletes  $m$  observations** from the original sample





A machine is used for packing grains in a sack. The weight of every bag is recorded as follows:

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 15.6 | 15.7 | 14.6 | 14.7 | 15.8 | 15.0 | 15.1 | 14.9 |
| 14.8 | 15.5 | 15.4 | 15.3 | 14.8 | 15.1 | 14.8 | 14.9 |

Draw at least the two samples and obtain their means. Compare this with the population mean.

# Sampling variability

- Let us draw four different samples from the data
- The sample mean varies from sample to sample
- This intuitive idea is called sampling variability
- The sample mean is close to the population mean

| Population Mean |
|-----------------|
| 15.125          |

| Sample                                         | Sample Mean |
|------------------------------------------------|-------------|
| 14.7, 15.1, 15.0, 15.6, 14.9, 15.4, 15.8, 15.1 | 15.20       |
| 15.8, 15.5, 15.3, 15.4, 15.8, 14.8, 14.6, 14.8 | 15.250      |
| 15.7, 15.6, 14.8, 14.8, 14.7, 14.9, 15.6, 15.1 | 15.150      |
| 14.8, 15.1, 15.7, 14.6, 15.6, 15.4, 15.6, 15.0 | 15.225      |

This file is meant for personal use by ravikanthamajala301@hotmail.com only.

Sharing or publishing the contents in part or full is liable for legal action.

# Parameter and Statistic

- Parameter:
  - The population quantities are termed as a parameter
  - In inferential statistics, sample is used to estimate the population parameters
  - Example: Population mean
- Statistic:
  - Statistical measure **computed** from sample observations
  - Also known as an estimator
  - Example: Sample mean

# Parameter and Statistic

The statistical notations for population parameters and sample statistics

|                    | Population Parameter | Sample Statistic |
|--------------------|----------------------|------------------|
| Mean               | $\mu$                | $\bar{x}$        |
| Variance           | $\sigma^2$           | $s^2$            |
| Standard Deviation | $\sigma$             | $s$              |
| Proportion         | $p$                  | $\bar{p}$        |

This file is meant for personal use by ravikanthamajala301@hotmail.com only.

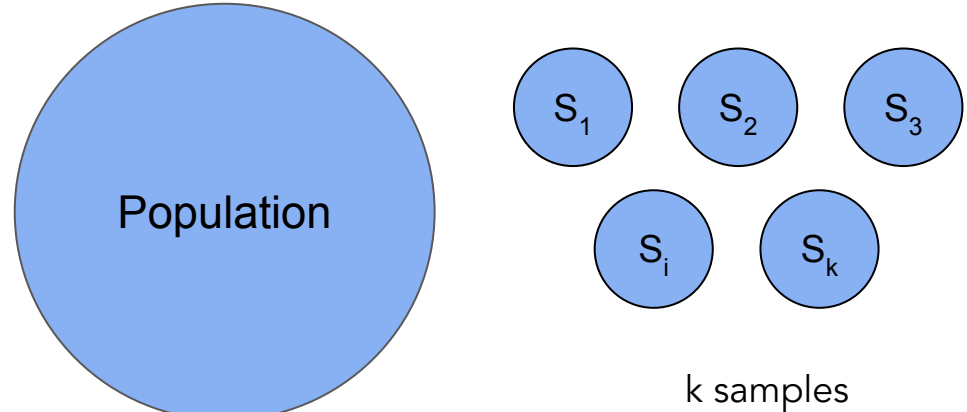
Sharing or publishing the contents in part or full is liable for legal action.



# Sampling Distributions

# Sampling distribution

- A total of  ${}^N C_n$  ( $= k$  say) samples can be drawn from a population of size  $N$  and sample size  $n$
- Obtain the mean and variance of each of these samples



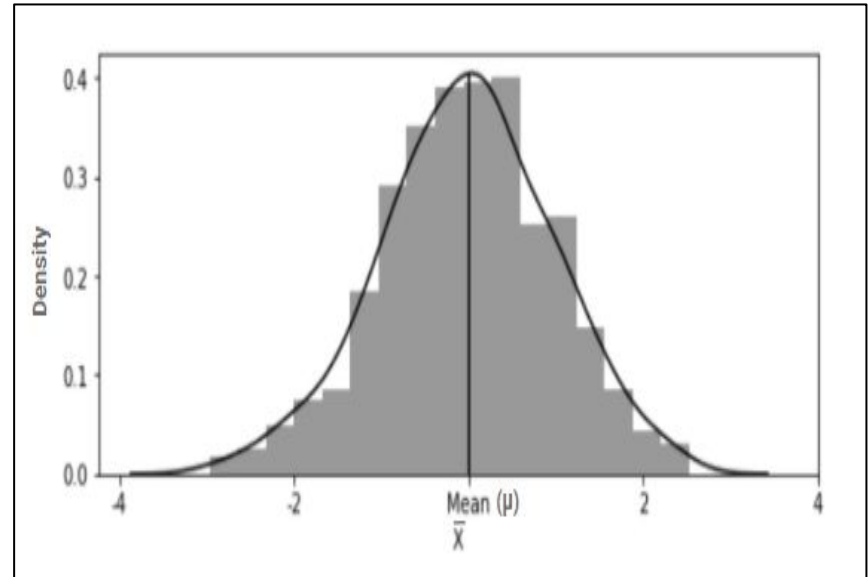
# Sampling distribution of sample mean

- The set of random sample means  $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_k$  are the observations for sampling distribution of sample mean
- These sample means follow the normal distribution
- Also, the mean of sample means is the population mean  $\mu_x = \mu$

| Sample No | Sample Mean |
|-----------|-------------|
| 1         | $\bar{x}_1$ |
| 2         | $\bar{x}_2$ |
| 3         | $\bar{x}_3$ |
| $\vdots$  | $\vdots$    |
| k         | $\bar{x}_k$ |

# Sampling distribution of sample mean

- The mean of sampling distribution of sample mean is  $\mu_{\bar{x}}$



## Central limit theorem

- The central limit theorem says that the sampling distribution of the mean will always be normally distributed, as long as the sample size is large enough.
- Regardless of whether the population has a normal, Poisson, binomial, or any other distribution, the sampling distribution of the mean will be normal.



## Central limit theorem

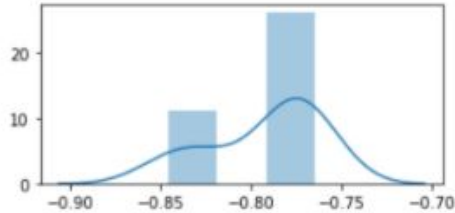
The central limit states that for random samples of large size  $n$  (usually  $n > 30$ ) taken from a single population with mean  $\mu$  and standard deviation  $\sigma$ , the sampling distribution of mean follows a normal distribution with mean  $\mu_{\bar{x}}$  and standard deviation  $\sigma/\sqrt{n}$  or  $s/\sqrt{n}$  which ever is known.

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

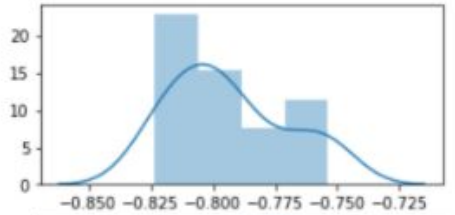


# Central limit theorem

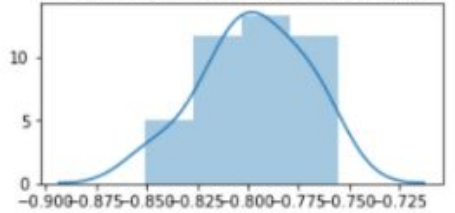
nSamples: 10



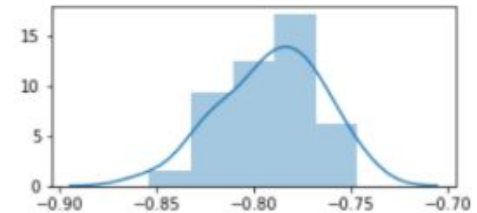
nSamples: 15



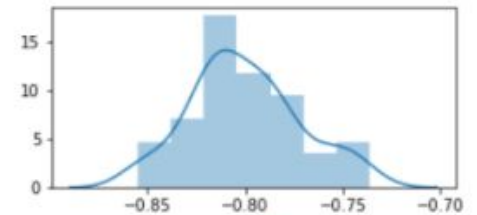
nSamples: 25



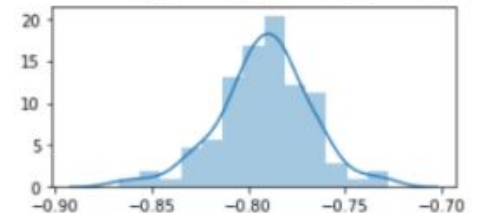
nSamples: 30



nSamples: 50



nSamples: 100



This file is meant for personal use by [ravikanthamajala301@hotmail.com](mailto:ravikanthamajala301@hotmail.com) only.

Sharing or publishing the contents in part or full is liable for legal action.

# Proportion

- Computed for a categorical variable
- Proportion (p) is the ratio of the number of observations of the desired category to the total number of observations

- It is given by 
$$p = \frac{x}{n}$$

where x is the number of observations of the desired category and n is the total number of observations

This file is meant for personal use by ravikantmamajala301@hotmail.com only.

Sharing or publishing the contents in part or full is liable for legal action.



# Sampling Distribution for Proportions

- The sample proportion is an estimate of the population proportion
- Like the sampling distribution of mean is obtained from samples, the sampling distribution for proportions can also be obtained
- The set of sample proportions are the observations for the sampling distribution of the sample proportion

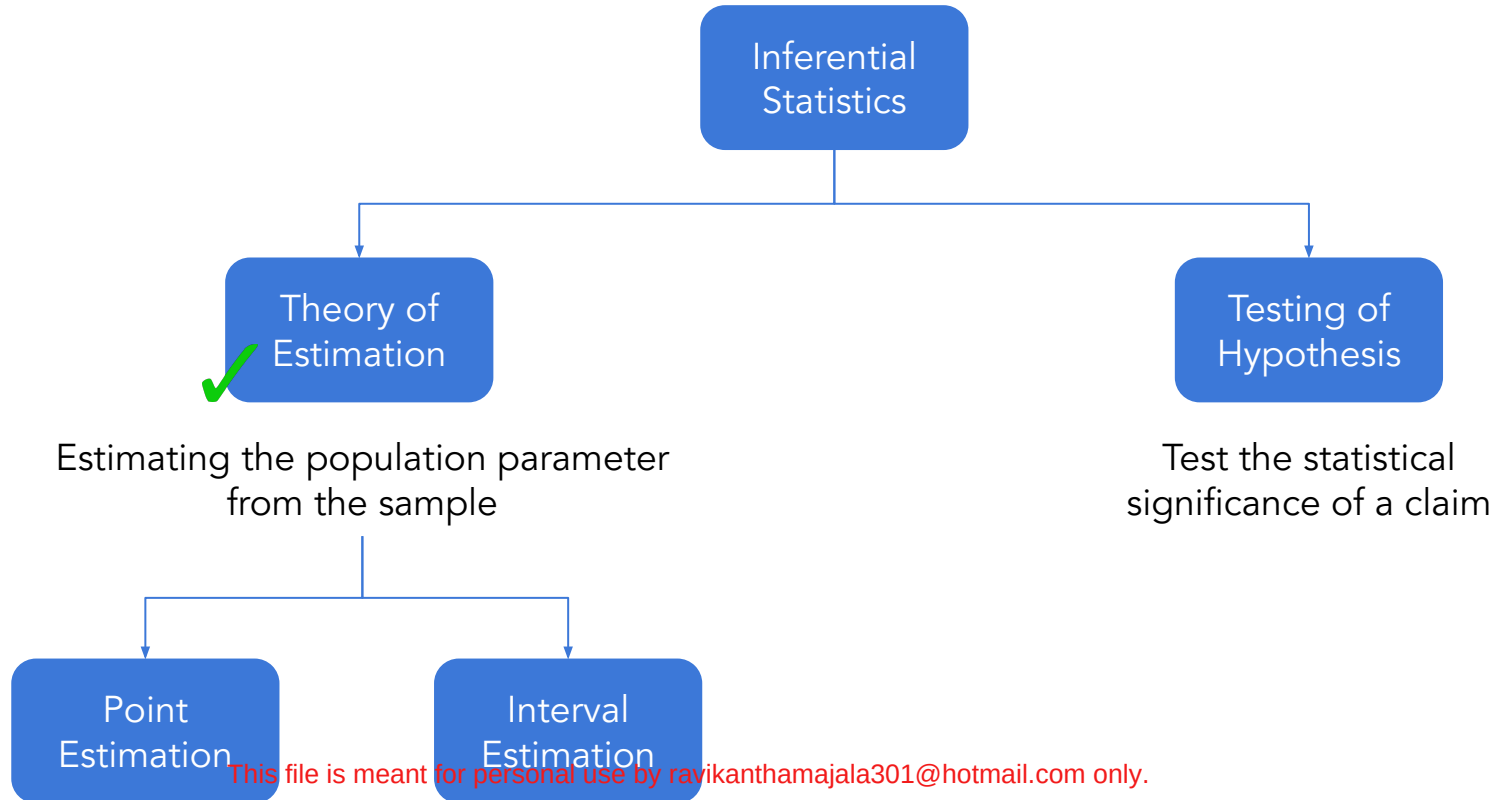
| Sample No | Sample Proportion |
|-----------|-------------------|
| 1         | $\bar{p}_1$       |
| 2         | $\bar{p}_2$       |
| 3         | $\bar{p}_3$       |
| $\vdots$  | $\vdots$          |
| k         | $\bar{p}_k$       |

# Summary

- The parameter describes the population and the statistic describes the sample
- There are different notations for population parameter and sample statistic
- Sampling distribution is the distribution associated with sample statistic
- Sampling distribution of the sample mean follows normal distribution with mean  $\mu_x$  and standard deviation  $\sigma/\sqrt{n}$
- Proportion is calculated for the categorical variable

# Inferential Statistics

# Inferential statistics



# Point Estimation



## Estimate

Question:

What would be the average income of a 27-year-old in London? How would you calculate it?



## Estimate

Solution:

The trivial way is to collect the income data of every 27-year-old, further compute its mean.

This mean would be the population mean.

# Point estimation

Is this scenario feasible? Is it possible to collect data of every 27-year-old person?

- To do so, draw a sample of size  $n$  and compute the mean.
- This sample mean is the point estimate of the population mean

This file is meant for personal use by [ravikanthamajala301@hotmail.com](mailto:ravikanthamajala301@hotmail.com) only.

Sharing or publishing the contents in part or full is liable for legal action.



# Point estimation

- The sample estimate is considered to be the point estimate of the population parameter
- If population parameters are not known they are estimated from the sample
- The point estimate of
  - population mean (  $\mu$  ) is sample mean (  $\bar{X}$  )
  - population variance (  $\sigma^2$  ) is sample variance  $s^2$
  - Population proportion (P) is sample proportion (p)



## Point Estimate

Question:

A financial firm has created 50 portfolios. From them a sample of 13 portfolios was selected, out of which 8 were found to be underperforming. Estimate the number of underperforming portfolios?



This file is meant for personal use by [ravikanthamajala301@hotmail.com](mailto:ravikanthamajala301@hotmail.com) only.

Sharing or publishing the contents in part or full is liable for legal action.



## Point Estimate

Solution:

A financial firm has created 50 portfolios. Thus  $N = 50$

From them a sample of 13 portfolios was selected. Thus  $n = 13$

The desired category is of underperforming portfolios. 8 of 13 were found to be underperforming. Thus  $x = 8$

To find: Estimate the number of underperforming portfolios



## Point Estimate

Solution:

The estimate of proportion of underperforming portfolios is given by

$$\hat{p} = \frac{x}{n} = \frac{8}{13}$$

To estimate the number of underperforming portfolios multiply by the estimate by N.

$$\text{The number of estimated underperforming samples} = \hat{p} \cdot N = \frac{8}{13} \cdot 50 = 30.76 \approx 31$$



## Point Estimate

Python solution:

```
# total count of portfolios
N = 50

# number of portfolios in a sample
n = 13

# number of underperforming portfolios in a sample
x = 8

# sample proportion
p_samp = x/n

# estimate the number of underperforming portfolios
num_port = p_samp*N

# round the number to get an integer value
print('The number of underperforming portfolios:', round(num_port))

The number of underperforming portfolios: 31
```

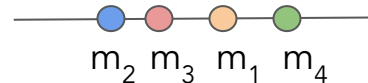
This file is meant for personal use by ravikanthamajala301@hotmail.com only.

Sharing or publishing the contents in part or full is liable for legal action.

# Demerits of point estimate

- In most of the scenarios, the point estimate is not accurate
- It is the single value representation of the population that differs with each sample
- The deviation of the point estimate from the population parameter gives rise to **sampling error**

Sample means of  
different samples



# Sampling error

- Sampling error is the absolute difference between the population parameter and its sample statistic (point estimate)
- Since the entire population is not considered as the sample, the values of mean, median, quantiles, and so on calculated on sample differ from the actual population values



## Sample size

Question:

The average temperature on a summer morning in Alaska is  $55^{\circ}\text{F}$ . The research student of climate studies records the temperature for 15 summer mornings in  $^{\circ}\text{F}$ .

Data = [49.8, 52.3, 51, 56.9, 54.8, 63, 58.2, 54.1, 50.4, 49.2, 47, 51.3, 43.5, 56, 55]

Find the sampling error for the average temperature.



# Interval Estimation

# Need of interval estimation

- The point estimate is taken from a particular sample
- For a different sample, we get a different estimate (sampling variability)
- This problem is resolved if we obtained the **interval estimate**



This file is meant for personal use by ravikanthamajala301@hotmail.com only.

Sharing or publishing the contents in part or full is liable for legal action.

# Interval Estimation

- Interval estimate is a range or an interval within which the parameter lies for a stated **confidence level**
- Based on the central limit theorem
- It is known as the confidence interval

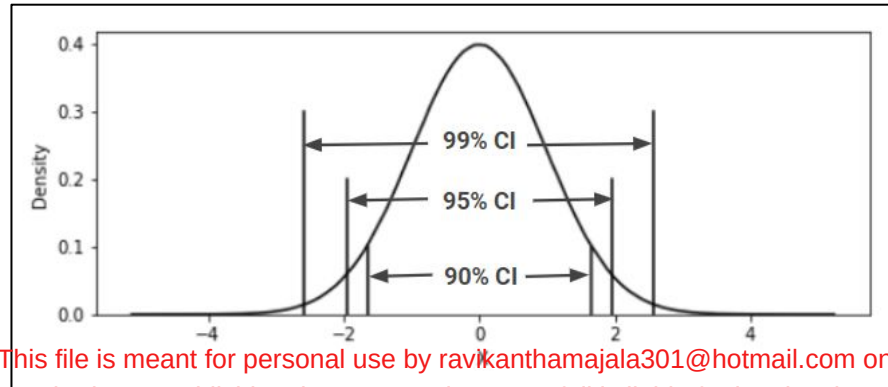
# Confidence level

- Confidence level ( $1-\alpha$ ) is the percentage of all possible point estimates that can be expected to include the actual population parameter
- $\alpha$  is known as the level of significance
- The higher confidence level illustrates the wider confidence interval
- A 95% confidence level implies that 95% of the confidence intervals would include the actual population parameter

# Confidence level

## Example:

The average income estimate for a 27-year-old in London is £45,000. After taking repeated samples, the 99% confidence intervals of the average income include the actual population parameter for 99% of the times.



This file is meant for personal use by ravikanthamajala301@hotmail.com only.

Sharing or publishing the contents in part or full is liable for legal action.

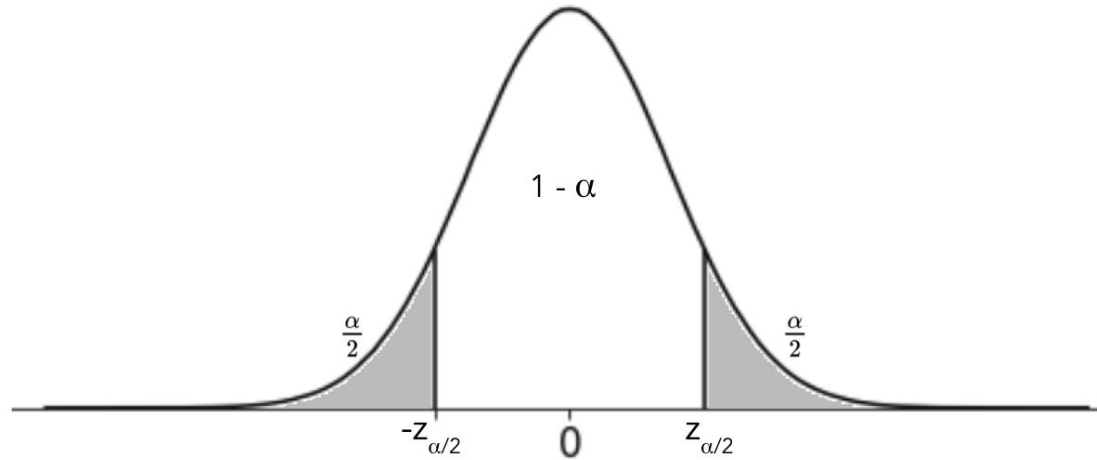
# Recall - CLT

For random samples of large size  $n$  taken from a single population with mean  $\mu$  and standard deviation  $\sigma$ , the sampling distribution of mean follows a normal distribution with mean  $\mu_{\bar{x}}$  and standard deviation  $\sigma/\sqrt{n}$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

# Interval estimate

From the CLT,  $P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$



Where  $\alpha$  is the level of significance

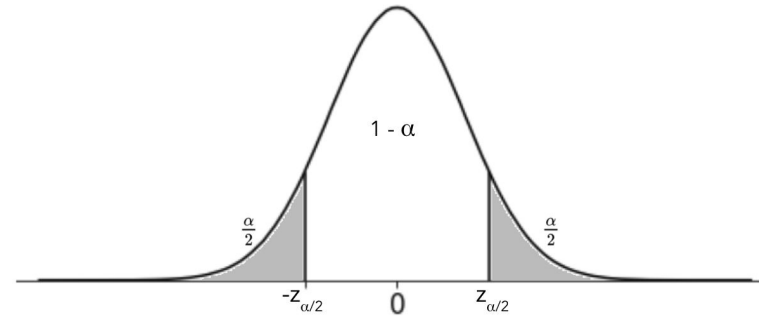
This file is meant for personal use by ravikanthamajala301@hotmail.com only.

Sharing or publishing the contents in part or full is liable for legal action.

# Interval estimate

From the CLT,  $P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$

| Value of $\alpha$ | $Z_{\alpha/2}$ | Value of $Z_{\alpha/2}$ |
|-------------------|----------------|-------------------------|
| 0.10              | $z_{0.05}$     | 1.645                   |
| 0.05              | $z_{0.025}$    | 1.96                    |
| 0.01              | $z_{0.005}$    | 2.575                   |



Note: To obtain these values in python use `scipy.stats.norm.isf()`



# Interval estimate

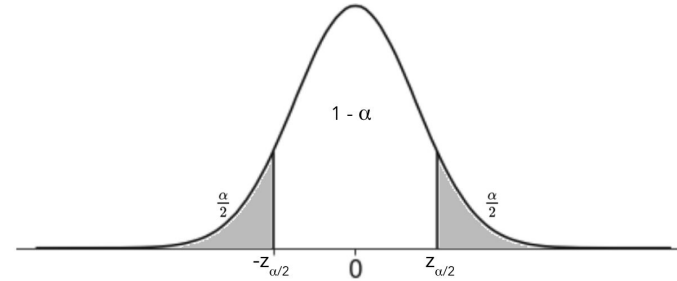
From the CLT,  $P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$

and value of  $Z$  is  $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$

Thus,  $P(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}) = 1 - \alpha$

$$\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Rearranging the terms,  $P(\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}) = 1 - \alpha$

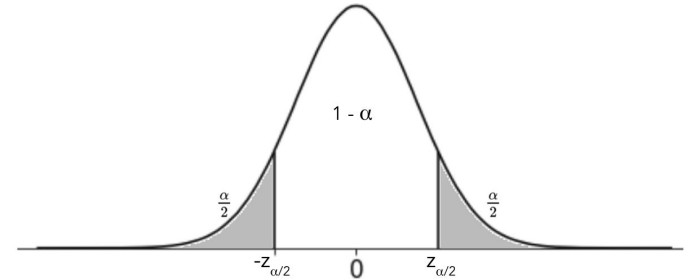


This defines  $(1 - \alpha)\%$  confidence interval

# Interval estimate

The  $100(1 - \alpha)\%$  confidence interval is

$$\underbrace{\bar{X}}_{\text{Point Estimate}} \pm \underbrace{z_{\alpha/2} \frac{\sigma}{\sqrt{n}}}_{\text{Margin of error}}$$





## Margin of error

Question:

100 bags of coal were tested and had an average of 35% of ash with a standard deviation of 15%. Calculate the margin of error for a 90% confidence level.



## Margin of error

Solution:

100 bags of coal were tested and had an average of 35% of ash with a standard deviation of 15%.

Here  $n = 100$

$$\sigma = 0.15$$

To find: The margin of error for a 90% confidence level.

Here  $\alpha = 0.10$ . Thus  $z_{\alpha/2} = 1.64$



## Margin of error

Solution:

The margin of error for a 90% confidence level is

$$z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 1.64 \frac{0.15}{\sqrt{100}} = 0.0246$$

A 90% CI with 2.46% margin of error implies that the sample mean will be within 2.46% points of the real population value 90% of the time



# Margin of error

Python solution:

```
# number of bags
n = 100

# standard deviation
std = 0.15

# given confidence level
conf_level = 0.90

# calculate z_alpha_by_2 with alpha = (1-conf_level) = 0.1
# use 'stats.norm.isf()' to find the Z-value corresponding to the upper tail probability 'q'
# pass the value of 'alpha/2' to the parameter 'q'
# use 'round()' to round-off the value to 4 digits
z_alpha_by_2 = np.abs(round(stats.norm.isf(q = 0.1/2), 4))

# calculate margin of error
error = (z_alpha_by_2*std)/ np.sqrt(n)

print('Margin of error:', error)

Margin of error: 0.024673499999999998
```

This file is meant for personal use by ravikanthamajala301@hotmail.com only

Sharing or publishing the contents in part or full is liable for legal action.



- As sample size increases the margin of error decreases
- If margin of error is fixed then the sample size can be obtained

$$E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$\Rightarrow n = z_{\alpha/2}^2 \left( \frac{\sigma^2}{E^2} \right)$$



## Sample size

Question:

For the previous question, for a margin of error 2.46. Verify whether you get sample size of 100.

Previous Question: 100 bags of coal were tested and had an average of 35% of ash with a standard deviation of 15%. Calculate the margin of error for a 90% confidence level.





## Interval estimate

### Question:

From a sample of 250 observations it is found that average income of a 27-year-old Londoner is £45,000 with a sample standard deviation of £4000. Obtain the 95% confidence interval to estimate the average income.



## Interval estimate

Solution:

A sample of 250 observations of a 27-year-old Londoner has an average income of £45,000 with a sample standard deviation of £4000

Thus,  $\bar{X} = 45,000$ ,  $\sigma = 4000$  and  $n = 250$

To find: The 95% confidence interval.

Here  $\alpha = 0.05$ . So  $z_{0.025} = 1.96$



## Interval estimate

Solution:

The 95% confidence interval is

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$\Rightarrow 45000 \pm 1.96 \frac{4000}{\sqrt{250}}$$

$$\Rightarrow (44504.155, 45495.845)$$



## Interval estimate

Python solution:

```
# number of observations in the sample
n = 250

# sample mean
sample_avg = 45000

# sample standard deviation
sample_std = 4000

# calculate the 95% confidence interval
# pass the sample mean to the parameter, 'loc'
# pass the scaling factor (sample_std / n^(1/2)) to the parameter, 'scale'
# as the population standard deviation is unknown, use the sample standard deviation
interval = stats.norm.interval(0.95, loc = sample_avg, scale = sample_std/np.sqrt(n))

print('95% confidence interval for average income:', interval)

95% confidence interval for average income: (44504.16397415635, 45495.83602584365)
```

This file is meant for personal use by ravikanthamajala301@hotmail.com only.

Sharing or publishing the contents in part or full is liable for legal action.



- It **cannot** be said that  $P(44504.155 \leq \mu \leq 45495.845) = 0.95$
- The population parameter  $\mu$  is a constant
- It **can** be said that from many samples drawn of size 250, construct a 95% CI for each sample, roughly 95% of the intervals would actually contain the population parameter  $\mu$



# Confidence Interval

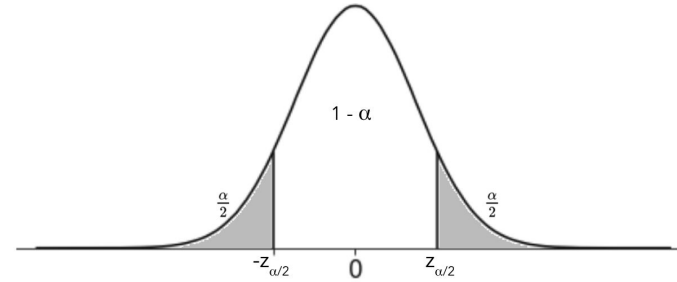
Question:

For the previous question, obtain the 90% and 99% confidence intervals. Further, logically explain which one of them is the widest.

# Interval estimate

The  $100(1 - \alpha)\%$  confidence interval for proportion is

$$\underbrace{\hat{p}}_{\text{Point Estimate}} \pm \underbrace{z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}_{\text{Margin of error}}$$





## Interval estimate

### Question:

A financial firm has created 50 portfolios. From them a sample of 13 portfolios was selected, out of which 8 were found to be underperforming. Construct a 99% confidence interval to estimate the population proportion.





## Interval estimate

Solution:

From them a sample of 13 portfolios was selected. Thus  $n = 13$

The desired category is of underperforming portfolios. 8 of 13 were found to be underperforming. Thus  $x = 8$

Thus the estimate of sample proportion is  $x/n = 8/13 = 0.615$

To find: Construct a 99% confidence interval to estimate the population proportion.



## Interval estimate

Solution:

Here  $\alpha = 0.01$ . So  $z_{0.005} = 2.575$

The 99% confidence interval is

$$\begin{aligned} & \hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \\ \Rightarrow & 0.615 \pm 2.575 \sqrt{\frac{0.615(1-0.615)}{13}} \\ = & (0.4381, 0.7926) \end{aligned}$$



## Interval estimate

Python solution:

```
# total count of portfolios
N = 50

# number of portfolios in a sample
n = 13

# number of underperforming portfolios in a sample
x = 8

# sample proportion
p_samp = x/n

# calculate the 99% confidence interval
# pass the sample proportion to the parameter, 'loc'
# pass the scaling factor ((p_samp*(1-p_samp))/n)^0.5 to the parameter, 'scale'
interval = stats.norm.interval(0.99, loc = p_samp, scale = np.sqrt((p_samp*(1-p_samp))/n))

print('99% confidence interval for population proportion is', interval)
```

99% confidence interval for population proportion is 0.262220927

# Summary

- Sample estimate of a population parameter is considered as a point estimate
- Sample mean, median, standard deviation are the examples of the point estimate
- Point estimate varies significantly for different samples
- Confidence interval returns the interval estimate for the population parameter
- The confidence level describes the uncertainty of the estimate
- Confidence interval is calculated using the point estimate and the margin of error
- The optimal sample size can be obtained by deciding the required margin of error

# Thank You